

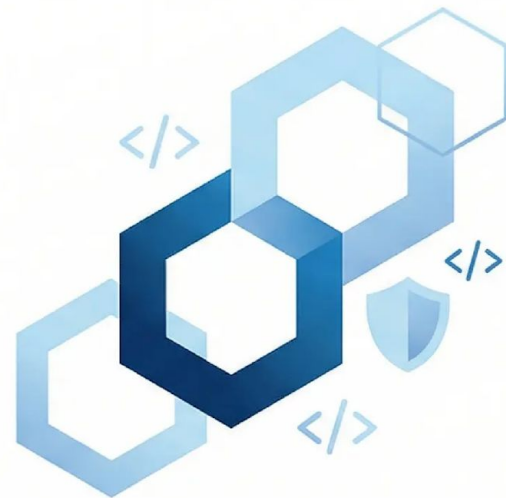
Open Source and Cybersecurity Fundamentals

A Comprehensive Learning Roadmap



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Founder, Bug2Build Community



About Me

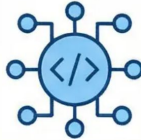
I'm Kastab Garai, a Cyber-Security researcher and Founder of the Bug2Build Community, a space that empowers hackers, builders, and students to explore technology.

Bug2Build is a professional space that empowers hackers, builders, and students to explore technology responsibly and ethically



Presentation Agenda

Part 1: Open Source Fundamentals



- Understanding Open Source & Licensing
- Global Developer Culture & Trends
- Monetization Models & Popular Tools

Part 2: Cybersecurity Career Roadmap



- Cybersecurity Fundamentals & Fields
- Step-by-Step Getting Started Guide
- Hands-On Learning & Certifications

What is Open Source?

Definition

Open source means software whose source code is publicly available and can be used, modified, and shared by anyone under specific conditions. This model promotes transparency, rapid innovation, and global collaboration.

Key Principles

Free Redistribution - Anyone can distribute the software

Source Code Availability - The source code must be accessible to users

Derived Works Allowed - Users can create and share modified versions

Integrity of the Author's Source Code - Modifications must be clearly marked

No Discrimination Against Persons or Groups
- No restrictions on individuals or groups

Distribution of License - License must be included when distributed

What is a License?



A license is a legal permission to use a copyrighted work.

- **It:**

- Grants permission to use the work
- Grants some or all copyright rights
- Can impose restrictions (usage type, location, duration, etc.)
- **Does NOT transfer ownership of copyright**

The license grants usage rights but ownership remains with the original copyright holder

Popular Open Source Licenses

MIT License

Highly permissive; allows reuse with minimal conditions

Apache License 2.0

Permissive with explicit patent grants; popular in corporate projects

BSD 3-Clause (Revised)

Simple, permissive license with attribution and no endorsement clauses

GNU General Public License (GPL) v2.0 / v3.0

Copyleft license; requires derivative works to remain open source

ISC License

Minimalist permissive license, similar to MIT

Mozilla Public License 2.0 (MPL 2.0)

Weak copyleft; allows linking with proprietary code while keeping modifications open

The Unlicense

Public domain dedication; no conditions, no restrictions

License Highlights

- Permissive Licenses (MIT, Apache, BSD): Ideal for broad adoption and integration into commercial software
- Copyleft Licenses (GPL): Ensure openness by requiring derivative works to be shared under the same terms

Open Source Culture: India vs United States



India

- Recently overtook the US in GitHub contributions
- 14% of all new GitHub users in 2025
- Largest contributor base globally
- Projected 57+ million developers by 2030
- 1 in 3 new coders worldwide



United States

- Traditionally largest open source contributor
- Strong job market and global influence
- Hosts major foundations (Linux Foundation)
- Major corporate contributors: Google, Microsoft, Cisco
- Leads in contribution per developer



India's rise reflects a global shift in developer talent

Global Open Source Growth



China

- Developer pool grew over 10× since 2015
- Major engine of global open source innovation
- Strong government and corporate support



Emerging Contributors

Brazil, Indonesia, Germany

- Rising number of contributors
- Vibrant local communities
- Increasing project creation
- Global shift toward inclusion



Per Capita Leaders

Switzerland & Nordic Countries

- Highest contributions per capita
- Small populations, big impact
- Often outperform larger nations

per capita

Global open source reflects a shift toward inclusive, worldwide development

How Do Licensors Make Money with Open Source Software?

Open source software is free, but companies earn revenue by providing additional services:



Support

24/7 technical assistance



Training

Workshops and certification



Customization

Tailored business solutions



Integration

Connecting existing systems



Certification

Validating skills and expertise



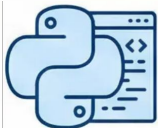
Warranties

Legal assurances for reliability

Key Insight: The software itself is free, but the value lies in additional services

Popular Open Source Tools & Languages

Open source powers modern technology



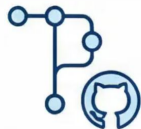
Python

A powerful scripting language for automation, data science, and web development



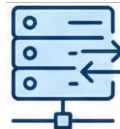
Tor Browser

Enables anonymous browsing and protects online privacy



Git & GitHub

Version control system for tracking code changes and collaborating on projects



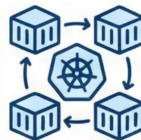
Apache

A widely used open-source web server for hosting websites and applications



VSCode

A free, open-source code editor based on Visual Studio Code, with no telemetry



Kubernetes

An open-source platform for automating deployment, scaling, and management of containerized applications



Chromium

The open-source browser project behind Google Chrome, powering modern web experiences

What is Cyber Security?

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks, unauthorized access, and damage.

It ensures:



Confidentiality

Only authorized users
can access data



Integrity

Data remains accurate
and unaltered



Availability

Systems and data are
accessible when needed

The term 'cyber' refers to computers, networks, and the internet. It comes from 'cybernetics,' related to communication and control in machines.



Introduction to Cyber Security Fields

Cybersecurity has many specialized domains that protect different aspects of digital systems

- Understanding these domains helps you identify where your interests and skills align
- Each field requires different technical skills and knowledge areas
- Specialization allows for deep expertise and focused career development

Let's explore the main cybersecurity fields...

Cyber Security Fields (Part 1)



Network Security

Protecting networks using firewalls, IDS/IPS, and VPNs



Application Security (AppSec)

Securing apps from vulnerabilities like SQL Injection and XSS



Information Security (InfoSec)

Protecting data through the CIA Triad (Confidentiality, Integrity, Availability)



Cloud Security

Securing AWS, Azure, and GCP environments



Endpoint Security

Protecting devices like laptops, servers, and mobile phones



Identity & Access Management (IAM)

Managing user authentication and authorization



Security Operations (SOC)

Monitoring systems and responding to threats in real time



Incident Response & Digital Forensics

Investigating breaches and collecting digital evidence



Penetration Testing (Ethical Hacking)

Simulating attacks to find and fix vulnerabilities



Malware Analysis

Studying malicious software to understand its behavior and impact

Cyber Security Fields (Part 2)

11



Cryptography

Encryption, hashing,
digital signatures

12



DevSecOps

Security in
development pipelines

13



Governance, Risk & Compliance (GRC)

Security policies and
regulations

14



Mobile Security

Android/iOS protection

15



IoT Security

Smart devices security

16



OT / Industrial Security

ICS, SCADA systems

17



Threat Intelligence

Analyzing threat actors
and risks

How to Start Cyber Security (Step 1-3)

1

Build Strong Foundations (0-3 Months)

- Learn internet basics: HTTP, DNS, TCP/IP
- Windows & Linux operating systems
- IP addresses, ports, and firewalls

2

Understand Cybersecurity Basics

- Master the CIA Triad: Confidentiality, Integrity, Availability
- Common types of cyberattacks (phishing, malware)
- Basics of encryption and data protection

3

Get Comfortable with Linux

- Install Kali Linux or VirtualBox
- Practice basic commands: file navigation, network configuration
- System monitoring fundamentals

These steps take just 0-3 months but lay the essential groundwork for a successful career in cybersecurity. Stay consistent!

Hands-On Learning

Practical experience is essential in cybersecurity. Theory alone isn't enough—real-world practice builds confidence and skill.

4

Learn Web Security

- Frontend ⇔ Backend ⇔ Database flow
- HTTP, cookies, and sessions
- Common vulnerabilities: SQL Injection, XSS, CSRF
- Master the OWASP Top 10

5

Start Hands-On Practice

Use safe, legal platforms:

- ✓ TryHackMe
- ✓ Hack The Box
- ✓ PortSwigger Web Security Academy

Practice these skills:

- Scanning
- Enumeration
- Ethical testing
- Reporting findings

⚠ Warning: Always practice on legal, authorized platforms. Never test systems without permission.

Choose Your Path



Offensive Security

Ethical hacking, penetration testing, vulnerability assessment



Defensive Security

SOC operations, threat detection, incident response



Cloud / DevSecOps

Secure pipelines, cloud hardening, container security



Governance (GRC)

Risk management, compliance, policy development

Tools & Certifications

Learn Tools Gradually



Nmap

Network scanning and discovery



Burp Suite

Web application security testing



Wireshark

Network traffic analysis



Metasploit

Penetration testing framework



OWASP ZAP

Automated security testing

Key Tip: Focus on understanding security concepts over memorizing tool commands. Tools are enablers – knowledge is the foundation.

Certifications (Optional but Helpful)

CompTIA Security+

Foundational security knowledge

Certified Ethical Hacker (CEH)

Offensive security & penetration testing

Google Cybersecurity Certificate

Beginner-friendly, hands-on learning

Microsoft Security Certifications

Cloud and enterprise security focus

THANK YOU

TOGETHER, WE CREATE IMPACT

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